

VIGHNAX — forecasting attacks before they complete

SIH 2026 · SIH26153 (NTRO) · AI-based Network Attack Forecasting from Network Traffic Data · Team VIGHNAX (127364)

Problem: IDS models label each flow in isolation and fire after the fact; an infiltration is a process unfolding over time.

Idea: learn a world model of network state dynamics $P(S_{t+1} \mid S_{\leq t})$ from flow + packet telemetry, and simulate K steps ahead.

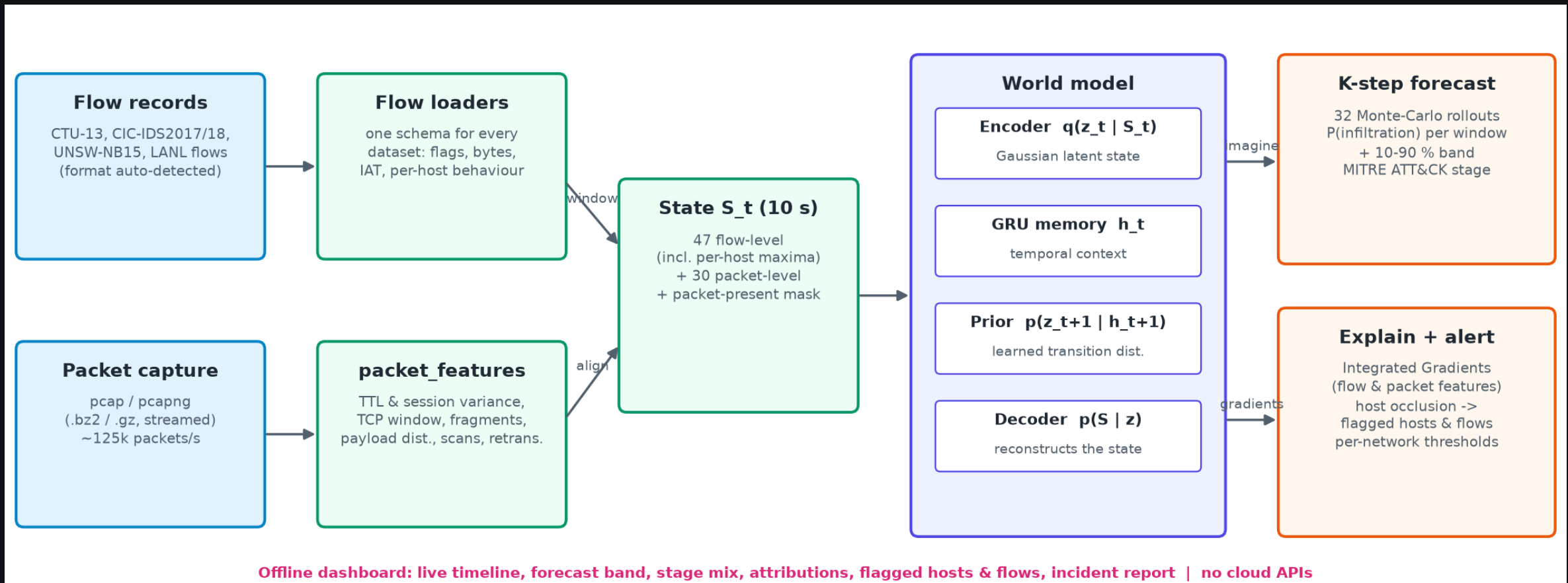
Output: P(infiltration in next 60 s) with an uncertainty band, predicted MITRE ATT&CK stage, driving features, flagged hosts/flows.

Evidence: one model, all 7 PS datasets, unseen test data: forecast macro AUROC 0.70 vs 0.67 (LR); 54/201 attack onsets forecast before they began (LR 17).

Practical: open source (MIT), fully offline: ~8 ms per 10 s window on a laptop CPU, 0.04 ms batched on GPU; ~125k packets/s ingestion.

Architecture: telemetry → network state → world model → forecast

any PS dataset format or PCAP · flow-level + packet-level features fused into S_t every 10 s (78 dims)



Flow level: 5-tuple, flags, bytes, packets, duration, IAT, fwd/bwd ratios + per-host maxima (a single bot is not diluted).

Packet level: TTL & session variance, TCP window, fragments, payload distribution, port-scan signatures, retransmissions.

World model: learn the dynamics, then imagine the future

RSSM-style latent state-space model (cf. World Models, Dreamer), trained on seven datasets' attack timelines

Transition model: encoder $q(z_t|S_t)$, GRU memory h_t , prior $p(z_{t+1}|h_{t+1})$ (learned transition distribution), decoder $p(S|z)$.

Supervised dynamics: attack + ATT&CK-stage heads trained on observed states AND on imagined K-step rollouts against the true future labels.

Forward simulation: 32 Monte-Carlo trajectories, $K = 6 \times 10$ s: probability per future window, 10–90 % band, stage mix.

Explainability: Integrated Gradients (Aumann–Shapley/SHAP family) vs a benign reference state; host-occlusion flags the responsible hosts.

Robustness: dataset-balanced training, modality dropout (CSV-only inputs), volume augmentation, per-site alert calibration.

Stages: Recon · Initial Access · Lateral Movement · C2 · Exfiltration · Impact — all present across the seven datasets.

Results: seven datasets, test data unseen in training

AUROC world model / logistic regression on the same features; thresholds calibrated on validation only

Dataset (unseen test data)	Forecast WM / LR	Early warning WM / LR	Detection WM / LR
CTU-13	0.83 / 0.60	0.82 / 0.48	0.89 / 0.74
CIC-IDS2017	0.56 / 0.56	0.47 / 0.47	0.65 / 0.65
CSE-CIC-IDS2018	0.83 / 0.60	0.57 / 0.55	0.83 / 0.62
UNSW-NB15	0.95 / 0.99	0.63 / 0.71	1.00 / 0.95
LANL cyber1	0.37 / 0.56	0.34 / 0.57	0.40 / 0.55
DARPA 1999	0.63 / 0.69	0.60 / 0.61	0.69 / 0.81
CICIoT2023	n/a	n/a	0.99 / 0.76
Macro average (temporal)	0.70 / 0.67	0.57 / 0.57	0.74 / 0.72

Forecasting: ahead on 2/6 datasets; early warning ahead on 2/6.

Before onset: 54/201 attacks forecast before they began (LR 17).

Honest: detection: LR better on LANL cyber1, DARPA 1999.

Note: CICIoT2023 episodes are composed, so detection only.

Demo, deployment and next steps

python server.py → <http://127.0.0.1:5000> (no internet needed)

Demo: drop any PS dataset file or PCAP (format auto-detected) or pick one of 8 bundled real captures → timeline, forecast fan, ATT&CK stage mix, IG attributions and flagged hosts & flows all update live as each window is processed.

Reproducible: fetch_datasets.sh → prepare_data.py / prepare_all.py → train.py → evaluate.py; seeded YAML config; weights + results in the repo; pytest suite.

Enterprise / CII fit: streams captures in bounded memory, CPU or GPU, fully offline, explains every alert, per-site threshold profiles.

Limitations: CICIoT2023 episodes are composed; LANL labels come from authentication-based red-team events; alert thresholds need per-site calibration.

Next: graph (host-level GNN) state, online threshold calibration, streaming NetFlow/IPFIX collector, SOC integration (syslog/STIX).